

Natural Extremolyte Fights Pigmentation Caused by Environmental Stressors

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Abstract

The natural extremolyte bitop Ectoin® natural shows comprehensive skin protection efficacy against multiple external stress factors initiating pigmentation, in this case particulate matter and blue light irradiation.

It effectively reduces pollution induced POMC gene expression and is capable to protect skin and skin cells from the whole spectrum of air pollution including PAHs, heavy metals as well as PM of all particle sizes. bitop Ectoin® natural also prevents blue light induced pigmentation and oxidative stress and additionally supports gentle skin lightening during long-term application. Furthermore, the natural active ingredient is supported by more than 15 years of published clinical data, proving its safety and efficacy for various applications including its therapeutic use in medical products.

Introduction

Human skin is situated at the interface of the organism and its environment and is therefore constantly exposed to numerous extrinsic stress factors. Chronic exposure to environmental conditions, including ultraviolet light (UV), blue light and air pollution has been shown to have significant impact to skin health and aging. It is an indisputable fact, that cutaneous aging is mainly driven by external stress factors, promoting the formation of reactive oxygen species (ROS), encouraging inflammation processes in cells and causing wrinkles as well as pigmentation.

Pigmentation of the skin results from the synthesis of melanin in the pigment-producing cells, the melanocytes, followed by distribution and transport of pigment granules to neighbouring keratinocytes. It is highly heritable, being regulated by genetic, environmental, and endocrine factors that modulate the amount, type, and distribution of melanins in the skin, hair, and eyes. In addition to its roles in camouflage, heat regulation, and cosmetic variation, melanin is crucial for absorption of free radicals that have been generated within the cytoplasm by UV, and it acts as a direct shield from UV and visible light radiation [1, 2, 3]. Being the largest organ of the body that is always under the influence of internal and external factors, the skin often reacts to those agents by modifying the constitutive pigmentation pattern. These hyperpigmented lesions are called age spots, liver spots, lentigo senilis, or actinic lentigines.

The most common cause of hyperpigmentation is more likely a post-inflammatory response to light-induced skin damage [4, 5]. That response may be the result of an acute inflammatory event such as sunburn or repeated suberythral exposures to UV or blue light.

Protecting the skin from external stress-factors such as pollution, UV and blue light and the prevention of cell damage

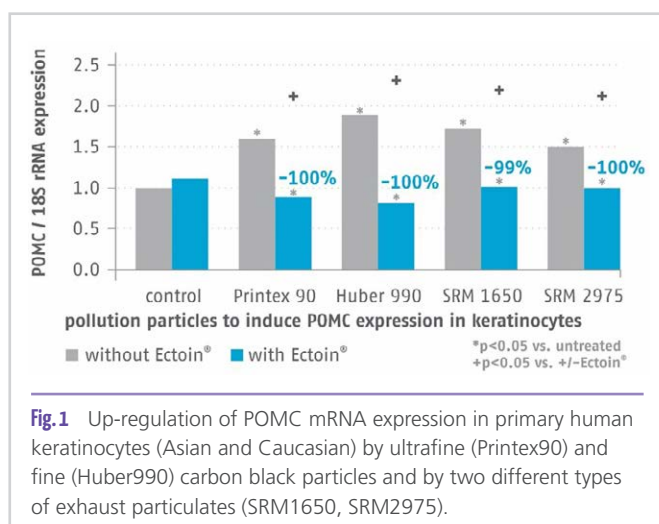
thus are logical and effective strategies to avoid hyperpigmentation.

In recent years, stress-protection molecules, so called extremolytes became popular as cosmetic active ingredients, e.g. bitop Ectoin® natural. Extremolytes are natural protection substances of extremophilic microorganisms and plants, which thrive in the most hostile environments our world provides – deserts, arctic ice, hot springs or the deep sea. Some halophilic bacteria accumulate the extremolyte bitop Ectoin® natural, to protect their cytoplasmic biomolecules against harmful environmental conditions like UV-light, heat, freezing, dryness, and osmotic stress [6].

bitop Ectoin® natural is a highly water-binding amino acid derivate with various positive effects for human skin including the reduction of wrinkles, restoring and stabilizing the skin barrier or the protection of skin-cells from the negative impact of external stress factors.

Protection against Pollution Induced Pigmentation

In the past years, several epidemiological and mechanistic studies concluded a connection between pollutants and accelerating wrinkles as well as pigment spot formation. In some highly polluted areas like Beijing or New Delhi, air-pollution is most probably the main factor for premature skin aging, as pollutants and scattering particulates in the troposphere reduce the effects of shorter wavelength UVR and significant reductions in UV irradiance have been observed in polluted, urban areas. Air pollutants may not only induce skin aging but are also linked to causing or worsening acne, atopic dermatitis, eczema, allergic reactions or even skin cancer [7].



The first study, showing that particulate matter (PM) has a negative impact on human skin was conducted by Vierkötter *et al.* in 2010. In this study with 400 Caucasian women, the association between PM exposure and pigment spot formation was shown [7]. PM is a carrier for organic compounds like heavy metals and PAHs. PAHs are highly lipophilic and thus easily penetrate the skin. PAH and PM are known to modify the expression and the release of POMC (pro-opiomelanocortin) resulting in the formation of pigment spots. The clinical relevance of these scientific findings, showing the damaging impact of air pollution on human skin, touches upon aspects of both, prevention and treatment.

One effective strategy to protect skin from air pollution is to maintain a healthy epidermal barrier function. Larger pollution particles like PM₁₀ and PM_{2.5} can thus be prevented from penetrating the skin. But given that particulate matter consists to 80% of ultrafine particles (UFP) like PM_{0.1} and smaller [8] and the highest level of concentration of toxic compounds, like PAHs and heavy metals is related to ultrafine particles, anti-pollution strategies should particularly target ultrafine particles.

An *ex vivo* study with fresh epidermal keratinocytes from a female Caucasian and Asian donor has shown, that bitop Ectoin® natural is capable to protect skin cells against the damaging impact of pollution particles of various sizes, particularly from ultrafine particles. Cells were untreated and pre-treated (24h) with 2 mM bitop Ectoin® natural solution. Afterwards, cells were stressed with fine and ultrafine carbon black particles and different surrogates for authentic street particulate matter such as SRM 1650 and SRM 2975. After the particle stress, the expression of POMC mRNA was measured in keratinocytes by using real time PCR.

The results in **Fig. 1** show that fine and ultrafine carbon black particles and diesel particulate matter induced POMC mRNA expression. POMC is known for melanogenesis stimulation in human melanocytes and to cause dark spot formations. It can therefore be used as a marker gene for pigmentation [9]. Keratinocytes, protected with bitop Ectoin® natural significantly down regulated PM induced over-expression of POMC mRNA in all tested cases by 100% or by 99% (**Fig. 1**).

These study results have been confirmed in a recently completed placebo-controlled *ex vivo* study on human living skin explants where bitop Ectoin® natural demonstrated a protection efficacy against pollution induced pigmentation [10].

Protection against Blue Light Induced Pigmentation

The visible light spectrum responds to wavelengths from about 390 to 700 nm and includes blue light (450 to 495 nm) which is also emitted by digital devices like PC screens.

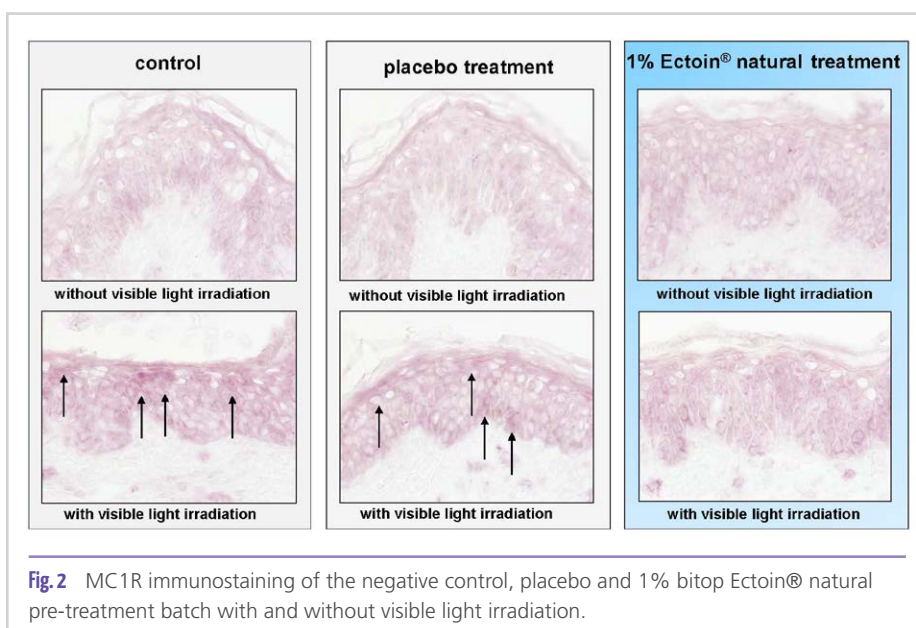
Various studies have shown negative effects to be exerted by visible light radiation including erythema, pigmentation, thermal damage and free radical production. Visible light can also induce indirect DNA damage due to the generation of reactive oxygen species (ROS) [11].

Most of the currently available UV-filters, organic and inorganic, offer, if any, weak protection against visible light. This limited effectiveness indicates the need for further protection concepts.

Botta *et al.* (2008) demonstrated in an *in vitro* study set-up that bitop Ectoin® natural prevents oxidative damage in the skin induced by visible light irradiation by up to 90% [12].

In an *ex vivo* study, human living skin explants were pre-treated with 1% bitop Ectoin® natural and exposed to visible light (65 J/cm²). Afterwards the parameters Nrf2 and MC1R were evaluated by immunostaining.

Nrf2 is a key transcription factor in the cellular response to



oxidative damage induced by reactive oxygen species (ROS). Nrf2 content in the cell increases automatically, when the skin is exposed to visible light or other stress factors. The study demonstrated the reduction of the cell's stress response to oxidative stress induced by visible light irradiation due to bitop Ectoin® natural pre-treatment. The presence of bitop Ectoin® natural decreased the need of Nrf2 in the cell. The results indicate, that bitop Ectoin® natural can protect the skin from visible light induced skin damage and photo-aging. Visible light has been reported to induce both, transient [13] as well as long lasting pigmentation in human skin [29]. MC1R is a marker for skin pigmentation and its activation leads to eumelanin production. Visible light increases MC1R content, which consequently leads to pigmentation of the skin. The results of the latest ex vivo study (Fig. 2) indicate a reduction of visible light induced melanogenesis due to bitop Ectoin® natural pre-treatment.

Whitening and Lightening Efficacy

Skin whitening or lightening remains being a trend in the cosmetic industry and is set to boost the global market. In general, skin whitening refers to the practice of using chemical substances to lighten the skin tone or to provide an even skin complexion by lessening the concentration of melanin. To accomplish this, there are several possible modes of actions suitable including the inhibition of tyrosinase activity, scavenging the intermediate products of melanin syntheses, preventing the transfer of melanosomes to keratinocytes, directly destroying existing melanin or destroying melanocytes. In the past, some countries banned unsafe, or downright hazardous skin lightening ingredients used in whitening products. Fortunately, science has discovered a number of natural whitening ingredients that work just as well as synthetic or chemical ones, without the harmful side effects. These natural ingredients work by absorbing harmful UV rays from the sun or inhibiting the production of melanin in the skin which

causes dark pigmentation. The demand for natural whitening actives is enormous, therefore an in vivo study was conducted to show the whitening and skin lightening efficacy of bitop Ectoin® natural.

In this study, a cream containing 1% bitop Ectoin® natural was applied on the face of the test persons twice daily for 84 days. All 22 test persons (Asian females, 36-60 years old) presenting photo type skin III and IV and at least one pigment spot with a diameter ≥ 3 mm on the face. The skin color was measured on day 0, 56 and 84 by Spectrocolorimeter® to define two different parameters: a depigmenting effect and the reduction of the different pigmentations.

bitop Ectoin® natural significantly increased the skin lightness by up to 3% and significantly decreased skin pigmentation by up to 28% (Tab. 1). These effects were observed on 100% of the test persons.

Furthermore the variation between pigmented skin and normal skin significantly decreased by up to 22% by the use of bitop Ectoin® natural (Tab. 2).

The tested subjects defined their satisfaction for the properties and efficacy of applying 1% bitop Ectoin® natural in a subjective questionnaire. 100% of the test persons stated that the skin was lighter after 84 days of application.

Conclusion

On the basis of its cell-protection benefits bitop Ectoin® natural offers whitening and lightening efficacy for the application in any kind of skin care concepts. In comparison to chemical substances bitop Ectoin® natural protects the skin from pigmentation in a very gentle manner. It reduces oxidative stress and damaging, as well as inflammatory gene expression only by mechanical stabilization of the cell membrane. Therefore, it is also perfectly suitable for long-term use. The anti-pollution efficacy of bitop Ectoin® natural has also been demonstrated in vivo in human lungs for therapeutic and preventative applications.

In conclusion, the natural extremolyte bitop Ectoin® natural is a safe and effective active ingredient, very well-suited for cosmetic formulations for all skin types, which need to match various consumer demands and market trends.

References

[1] Pathak, M. A., & Fanselow, D. L. (1983). Photobiology of melanin pigmentation: dose/response of skin to sunlight and its contents. *Journal of the American Academy of Dermatology*, 9(5), 724-733.

[2] Riley, P. A. (1997). Melanin. *The international journal of biochemistry & cell biology*, 29(11), 1235-1239.

[3] Bykov, V. J., Hemminki, K., & Marcusson, J. A. (2000). Effect of constitutional pigmentation on ultraviolet B-induced DNA damage in fair-skinned people. *Journal of investigative dermatology*, 114(1), 40-43.

[4] Gilchrist, B. A. (1998). The photobiology of the tanning response. The pigimentary system physiology and pathophysiology, 359-372.

[5] Abdel-Malek, Z., & Kadekaro, A. L. (2006). Human pigmentation: its regulation by ultraviolet light and by endocrine, paracrine, and autocrine factors. The pigimentary system: physiology and pathophysiology, 410-420.

[6] Lippert, K., & Galinski, E. A. (1992). Enzyme stabilization by ectoine-type compatible solutes: protection against heating, freezing and drying. *Applied microbiology and biotechnology*, 37(1), 61-65.

Parameter	Change in % after 56 days	Change in % after 84 days
Skin lightness	+ 2%	+ 3%
Skin pigmentation	– 25%	– 28%

Tab.1 Depigmenting effect and skin lightening.

Parameter	Change in % after 56 days	Change in % after 84 days
Variation between spot and normal skin	– 11%	– 22%

Tab.2 Reduction of different pigmentations.

- [7] Vierkötter, A., Schikowski, T., Ranft, U., Sugiri, D., Matsui, M., Krämer, U., & Krutmann, J. (2010). Airborne particle exposure and extrinsic skin aging. *Journal of investigative dermatology*, 130(12), 2719-2726.
- [8] WHO working group (2003). Health aspects of Air pollution with Particulate matter, Ozone and nitrogen dioxide. Bonn, Germany: Regional Office for Europe
- [9] Krutmann, J., Liu, W., Li, L., Pan, X., Crawford, M., Sore, G., & Seite, S. (2014). Pollution and skin: From epidemiological and mechanistic studies to clinical implications. *Journal of dermatological science*, 76(3), 163-168.
- [10] IUF – Leibniz Research Institute for Environmental Medicine (2018) Unpublished study. Düsseldorf, Germany.
- [11] Godley, B. F., Shamsi, F. A., Liang, F. Q., Jarrett, S. G., Davies, S., & Boulton, M. (2005). Blue light induces mitochondrial DNA damage and free radical production in epithelial cells. *Journal of Biological Chemistry*, 280(22), 21061-21066.
- [12] Botta, C., Di Giorgio, C., Sabatier, A. S., & De Méo, M. (2008). Genotoxicity of visible light (400–800 nm) and photoprotection assessment of ectoin, L-ergothioneine and mannitol and four sunscreens. *Journal of Photochemistry and Photobiology B: Biology*, 91(1), 24-34.
- [13] Pathak, M. A., Riley, F. C., & Fitzpatrick, T. B. (1962). Melanogenesis in human skin following exposure to long-wave ultraviolet and visible light. *Journal of Investigative Dermatology*, 39(5), 435-443.

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